Gaps in the Application of Network Forensics Tools Within the Automotive Industry

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Abstract

The modern vehicle industry grows more reliant on computerized infrastructure every year, and that dependence on digital embedded features demands specialized maintenance, upkeep, and protection in keeping with ever-changing standards. Since the turn of the century, all new vehicles sold in the United States have been required to carry an On-Board Diagnostics (OBD) computer that self-diagnoses and reports on critical systems and subsystems. That requirement alone is enough to consider most standard vehicles an Internet of Things (IoT) device: as a collection of smaller computers exchanging near-real-time data over mobile infrastructure, the modern automobile fits the technical definition.

While attacks on a vehicle's network capabilities are currently rare, the forensic analysis and reporting process for the few events that do occur is severely lacking. An entire car can become a near-two-ton, fast-moving weapon representing the attack vector for a malicious, computer-savvy individual — and today, not much useful evidence would be recovered afterward. The advent of autonomous vehicles will only widen this gap. Digital forensics will become more prominent within the automotive industry, but analysis techniques will have to catch up to the implementation gaps we currently face.

Introduction

Automotive network forensics is a branch of digital forensics that focuses on investigating network attacks and security incidents in modern vehicles. As today's vehicles become increasingly computerized and connected to the internet, they grow more vulnerable to a wide range of cyber threats, including remote hacking, malware infection, and data theft. The practice primarily involves the collection, preservation, and analysis of digital data from various components of the vehicle's computer network. Sources such as the electronic control units (ECUs), infotainment systems, and communication modules can provide critical detail about the nature of an attack, the identity of the attacker, and the extent of the damage (Gomez Boquerón, 2021).

The key challenges include the need for specialized tools and expertise, the difficulty of accessing and preserving data from ECUs, and the rapidly evolving nature of the threats themselves. These systems are now essential to the safe and efficient operation of newer vehicles, so any compromise of their security can carry serious consequences — beginning with loss of control and theft of personal data, and ending with physical harm to passengers or other road users.

Current State of the Industry

Automotive network forensics is still in its infancy, with much research and development needed before effective investigation techniques become standard. Some research has been conducted, but a lack of standardization and best practices makes it difficult to run consistent, reliable investigations across different types of vehicles.

One of the major challenges is the diversity of network protocols used across manufacturers. Different automakers may use different protocols, which frustrates any universal approach to forensics. In “A Generalized Approach to Automotive Forensics,” the authors describe a complex field, driven by the sheer variety of systems, devices, and technologies in modern vehicles. They argue that the lack of standardization in automotive network protocols, combined with the complexity of the underlying systems, makes forensic investigation particularly hard. They propose a generalized approach that spans data acquisition, analysis, and reconstruction, with a strong emphasis on identifying and preserving evidence (Gomez Boquerón, 2021).

A second challenge is the complexity of the network architecture inside modern vehicles compared to traditional environments. Some manufacturers rely on proprietary protocols, which makes a consistent forensic approach even harder to achieve. The growing number of networked devices within a vehicle enlarges the attack surface, and the interconnected nature of those devices makes it more difficult to isolate and investigate any single attack. The rapid evolution of threats compounds all of this, keeping investigators in a constant race to match current attack methods with adequate defenses.

Tools and Techniques

Automotive network forensics draws on a wide range of tools and techniques to investigate cyberattacks and other incidents on in-vehicle networks.

Network traffic analysis is foundational. It involves capturing and analyzing the traffic generated by the various ECUs in the vehicle. Tools such as Wireshark and tcpdump are common here, letting investigators analyze traffic in real time or work from captures saved for later. Traffic analysis can help identify the source of an attack, the types of data that were stolen, and the extent of the damage to the vehicle's systems.

Memory analysis examines the contents of the ECUs' memory to surface evidence of malicious activity or other anomalies. Tools such as Volatility and Rekall let investigators inspect memory contents and identify behavioral patterns that may indicate an attack (Diaz Linares, 2022). Memory analysis can reveal which malware was used and how it evaded the vehicle's security systems.

Data extraction pulls data directly from the vehicle's ECUs. Tools such as CarShark — a CAN-bus packet sniffer and injector developed for early automotive security research — allow

investigators to capture and analyze ECU traffic for signs of an attack (Ohira, 2023). Effective extraction can identify what data was stolen and the methods used to exfiltrate it.

Physical analysis examines the vehicle's physical components for evidence of tampering or manipulation. Techniques such as chip-off analysis (removing flash memory for direct examination) and, where warranted, microscopic inspection let investigators identify which components were targeted and how they were damaged or altered.

Reverse engineering rounds out the toolkit. A recovered vehicle may have its firmware or software reverse-engineered to surface evidence of malicious activity. Tools such as IDA Pro and Ghidra let investigators analyze ECU code, identify patterns indicative of an attack, and begin developing patches (Shin, 2022). Reverse engineering can also reveal which malware was used and how it avoided detection.

Emerging Trends

Automotive network forensics is evolving quickly, and several trends are shaping its future.

Machine learning and artificial intelligence. These technologies can analyze the large volumes of data generated by modern vehicles quickly and accurately, allowing investigators to identify suspicious patterns at a rate manual analysis can't match. Machine learning algorithms can be trained to flag anomalous behavior on a vehicle network that may indicate a cyberattack, and AI can automate parts of the investigation, such as surfacing patterns relevant to a case (Rizvi, 2022).

Cloud-based forensics. As vehicles become more connected, the ability to collect and analyze data from many sources at once grows more important. Cloud-based forensics lets investigators centrally store and analyze data across multiple vehicles, making it easier to detect patterns and identify vulnerabilities, and it supports real-time analysis for faster response (Jo, 2022). It also enables remote access to vehicle data — including cloud-backed services tied to infotainment, GPS tracking, and remote diagnostics — without physically accessing the vehicle.

Blockchain. A more divisive trend, blockchain can be used to store vehicle-generated data in a tamper-proof, immutable ledger (Mansor, 2016). This is appealing in investigations where data integrity is paramount, since it provides an additional layer of authentication for collected evidence. Its practicality in this setting remains contested.

Integrated hardware/software and digital twins. As automotive networks grow more complex, integrated solutions that combine hardware and software can provide more comprehensive network analysis. Digital twin technology is particularly promising: a digital twin is a virtual replica of a physical system that can simulate a vehicle's network to identify vulnerabilities. A complete digital twin emulates all vehicle software, ECUs, and physical behavior (Jo, 2022), letting each development step be simulated and validated so problems can

be caught before real parts are produced. Investigators can use a twin to locate weaknesses and test fixes in a safe, controlled environment.

Case Studies

Real-world incidents illustrate both the challenges and the value of automotive network forensics.

BMW ConnectedDrive (2015). Security researchers found a vulnerability in BMW's ConnectedDrive system that allowed remote access to vehicle functions, including unlocking doors and starting the engine. BMW issued a security patch, but the incident underscored the need for stronger security in connected cars.

Jeep Cherokee (2015). Researchers demonstrated a car's internal network being remotely compromised, exploiting a flaw in the Uconnect infotainment system to take control of steering, brakes, and engine. Fiat Chrysler recalled 1.4 million vehicles and issued a software update. The incident became a landmark case for automotive security and forensics.

Tesla Model S CAN-bus hack (2016). A group of Chinese researchers remotely gained control of a Model S's brakes, steering, and other systems by exploiting a vulnerability in the vehicle's Controller Area Network (CAN) bus. The case highlighted both the fragility of in-vehicle networks and the need for forensic techniques capable of tracing such attacks.

Tesla Autopilot crash (2016). A Model S crashed into a tractor-trailer while in Autopilot mode, killing the driver. The National Highway Traffic Safety Administration (NHTSA) concluded the crash stemmed from the driver's over-reliance on Autopilot and failure to maintain control. The case raised lasting questions about the safety of autonomous driving technology and the need for better data collection and analysis in these events.

Toyota unintended acceleration (early 2010s). Toyota faced lawsuits and recalls tied to reports of unintended acceleration, attributed to a combination of mechanical and electronic faults, including accelerator-pedal issues and software concerns. The episode raised concerns about the safety of automotive electronics and the adequacy of testing and validation.

Uber autonomous vehicle fatality (2018). An autonomous Uber test vehicle struck and killed a pedestrian in Tempe, Arizona. Investigations by the National Transportation Safety Board found that the vehicle's sensors detected the pedestrian several seconds before impact, but the perception system failed to classify her correctly — cycling among object types — and emergency braking was suppressed. The case is a defining example of why automotive network forensics is needed to reconstruct autonomous-vehicle incidents and identify the software or hardware faults that contributed.

Volkswagen emissions scandal (2015). Volkswagen was found to have installed software in its diesel vehicles that detected emissions testing and reduced output accordingly, while real-world

driving produced nitrogen-oxide emissions far above the legal limit. The case shows how forensic analysis of vehicle software can expose fraud and illegal activity in the industry.

Two additional investigations illustrate the field's breadth. In one, investigators used a vehicle's internal network data to determine that a driver had been using a cell phone moments before a fatal crash, establishing distraction as the primary cause. In another, forensic data from a stolen connected car's internal network was used to trace its location, driving behavior, and destination, leading to recovery of the vehicle and apprehension of the thief. Together these cases show the value of automotive network forensics in reconstructing events, determining liability, improving road safety, and supporting law enforcement.

Legal and Ethical Considerations

Privacy is a central concern whenever digital evidence is involved. Data collected during an investigation may contain sensitive personal information — location data, call records, text messages — that the owner never consented to disclose. Examiners should protect that data and share it only with authorized parties, and where possible obtain the vehicle owner's consent so the investigation proceeds lawfully.

Digital evidence must also be prepared for potential admissibility in court. That requires a strict chain of custody and clear, thorough documentation of every procedure and finding (Nilsson, 2010), so the integrity of the data is preserved and tampering is discouraged. Examiners must often be ready to explain and defend the methods they used to collect and analyze evidence — and to do so without letting invasive investigative work cause further damage to the vehicle's systems. All of this serves to protect individual privacy while keeping the collected evidence usable in court.

The Future

As the automotive industry integrates more advanced technology, automotive network forensics will play a growing role in detecting and preventing cyberattacks.

Machine learning is the clearest near-term driver. Given the volume of data modern vehicles generate, ML can automate much of the analysis, letting investigators identify threats faster and more accurately by learning normal patterns and flagging anomalies. AI can further support intelligent security systems that detect and respond to attacks in real time (Baig, 2017).

Autonomous vehicles will introduce their own challenges. Self-driving systems rely heavily on machine-to-machine communication, which will demand new forensic tools and techniques, and the volume of data they generate will require new approaches to storage and analysis. As attacks designed specifically for autonomous vehicles emerge, researchers expect to lean on machine learning for threat intelligence — the automated collection, analysis, and dissemination of information about potential threats (Le-Khac, 2020) — so automakers can stay ahead of emerging attacks with proactive measures.

Blockchain may improve the security and integrity of vehicle data through a decentralized, tamper-proof ledger, helping ensure that data used in investigations is accurate and authentic. It could also harden vehicle networks by supporting secure communication and preventing unauthorized access, though its practicality here remains highly contested (Baig, 2017).

Finally, the regulatory landscape continues to evolve, and forensics will have to keep pace. This includes data-privacy regulations such as the EU's General Data Protection Regulation (GDPR), along with industry standards for cybersecurity and forensics. The current automotive-security standards were catalyzed in part by the 2015 Jeep Cherokee hack and developed by the International Organization for Standardization (ISO) alongside SAE International, producing ISO/SAE 21434 — a standard defining concepts and safety minimums for cyber threat analysis and risk assessment across development, production, servicing, and management. Notably, the standard does not define or address specific vulnerabilities or how to fix them; its authors judged that enumerating those, against ever-evolving attacker capability, would be futile (Le-Khac, 2020). Investigators will need to track these developments closely to stay compliant.

Conclusion

The growth of computer systems in modern vehicles has sharpened the need for automotive network forensics. This survey examined the current state of the field, its major challenges, the tools and techniques in use, and the emerging trends and future possibilities that will shape it. The legal and ethical dimensions cannot be set aside: as vehicles become more connected, adhering to established guidelines and principles only grows more important. As technology evolves and more vehicles come online, the field will continue to grow, with new tools and techniques — AI and ML chief among them — developed to keep pace with emerging threats. Automotive network forensics is ultimately critical to the safety and security of drivers and passengers alike, and essential to the continued growth of the automotive industry.

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